

Host:

Dear viewers, hello! Welcome to the online video interview program of the Shanghai Qingpu District Government Portal. I'm your host for today. Today, we will discuss and exchange ideas on topics like smart education and the application of information technology. We are also very honored to have invited Mr. Li Wei, Director of the Science and Information Development Center at Shanghai Zhujiajiao Middle School and Senior Teacher of Information Technology, to our studio today. Teacher Li, hello! Please greet our audience and introduce yourself.

Li Wei:

Hello everyone! I am Li Wei from Shanghai Zhujiajiao Middle School. I have been engaged in information science teaching and the application of educational information technology for over 20 years.

Host:

To start with, what do you think are the main aspects of educational informatization in schools?

Li Wei:

In my opinion, educational informatization mainly consists of the construction of foundational infrastructure, the provision of rich digital educational resources, the integration of intelligent information technologies into educational and teaching reforms, and the refinement of applications based on educational big data analysis.

The fundamental goal of educational informatization is to promote quality education and cultivate innovative talents. In general, the core of educational informatization lies in teaching informatization. Teaching is the central component of education, and teaching informatization means making teaching methods more technological, disseminating educational information digitally, and modernizing teaching approaches.

In 2022, the National Smart Education Public Service Platform was officially launched, marking a significant milestone in China's digital education strategy. Over 70 million teachers have activated its use nationwide. The platform encompasses nine major application scenarios—self-directed learning, teacher lesson preparation, dual-teacher classrooms, assignment activities, Q&A support, after-school services, teacher training, home-school communication, and regional management—creating an educational environment where everyone can learn anywhere and at any time, providing quality education opportunities for every learner.

Host:

How different do you think it would be to study in a school where educational informatization is widely implemented?

Li Wei:

We can imagine that in schools with widespread implementation of educational informatization, students' academic performance, extracurricular activity participation, research studies, sports, and even dietary habits could all be automatically collected. An intelligent platform could then analyze the data and create a digital profile for each student, providing an objective evaluation of their performance. For example, if the system collects daily dietary information, it could generate a report on the student's nutritional intake and make recommendations for improvement in the following week.

In sports, wearable devices can track data such as heart rate and step count to analyze students' physical activity levels and help teachers design targeted lessons. Additionally, information technology can facilitate cross-class teaching activities. Under the flexible course selection system, students can have individual schedules to meet their personalized learning needs. In smart classrooms, students can use devices like tablets to interact during lessons.

By leveraging smart education platforms, the system can automatically generate records of students' mistakes and suggest personalized exercises based on subject-specific knowledge maps. Furthermore, students' evaluations can include longitudinal tracking of their learning progress and a holistic assessment of moral, academic, physical, artistic, and labor values.

Host:

You mentioned smart education earlier. Could you explain more specifically what smart education is?

Li Wei:

Smart education is a broad concept. IBM first proposed the "Smarter Planet" strategy in 2008, outlining a vision to create a new world operation model using advanced information technology. Its goal was to make almost everything on Earth perceptible, interconnected, and intelligent. Derived from this vision, the ideas of smart education and smart learning emerged.

Education is inherently tied to the times. Throughout human history, education has constantly evolved to adapt to societal changes. Emerging information technologies like the internet, artificial intelligence, big data, and blockchain have driven profound societal transformation and are steering education toward smart education.

Professor Gu from East China Normal University, a member of the Ministry of Education's informatization expert group, believes that the essence of smart education lies in creating an ecological learning environment that integrates technology. By utilizing human-machine collaborative data intelligence, teaching expertise, and cultural wisdom, guided by principles like precision, personalization, optimization, collaboration, critical thinking, and creativity, smart education empowers teachers to use high-efficiency

teaching methods and enables learners to receive appropriate, individualized learning services. Simply put, the essence of smart education is to leverage human-machine collaboration to optimize the teaching process and foster the future paradigm of education that accelerates learner development.

Host:

Speaking of the internet, ChatGPT has recently gained a lot of attention, especially regarding its impact on education. What is your opinion on this tool?

Li Wei:

ChatGPT has indeed been trending lately, reaching 100 million active users within just two months of its launch. I tried asking ChatGPT about its impact on education, and its response was: "ChatGPT is a new machine learning technology characterized by speed, accuracy, and efficiency, accelerating the development of AI in education, marking a qualitative leap in this field. On one hand, it can provide instant question-answering capabilities, and on the other, it can be used for content creation, improving education quality, among other things."

From its own response, we can see the positive impact of AI on education. AI products are not new; for example, smartphones can interact via voice commands through app-based AI. However, compared to other AI products, ChatGPT is exceptional in solving open-ended questions. With just a minimal prompt, ChatGPT can create well-crafted and eloquent content.

Many students admit to using ChatGPT to write their papers. Teachers often struggle to distinguish whether an assignment was completed by the student or created using ChatGPT. Therefore, educators must rethink teaching and assessment methods to spark innovative change.

Ni Minjing, Deputy Director of Shanghai Municipal Education Commission, has stated that urgent education reforms are needed to address these challenges:

The primary aim of education should shift to cultivating individuals capable of independent thought and sound value judgments rather than merely acquiring specific knowledge.

Educational methods and strategies need significant adjustments. Instead of avoiding or fearing tools like ChatGPT, educators should embrace them to innovate teaching approaches collaboratively.

In my opinion, the rapid development of AI technology presents both challenges and opportunities. In the future, it's not AI that will take over your job but rather individuals who know how to wield AI effectively. Teachers can utilize advanced AI tools and techniques to revolutionize lesson planning, assignment evaluation, and support students' independent research, thereby allowing AI to enhance teaching practices.

Host:

Zhujiajiao Middle School is a benchmark school for the application of educational informatization in Shanghai. Can you tell us how the school incorporates "smart" teaching practices to enhance teachers' wisdom and help students become smart learners?

Li Wei:

So far, Shanghai has been the only national pilot area for comprehensive educational reform, shouldering the critical responsibility of piloting digital transformation in education. Digitalization is not only a national strategy but also a major strategic priority outlined in Shanghai's "14th Five-Year Plan." Based on its precise teaching applications, Zhujiajiao Middle School has actively contributed to the development of informatization benchmark schools in Shanghai. In June 2019, our school was officially approved by the Shanghai Municipal Education Commission as part of the second cohort of informatization application benchmark schools.

Currently, we have officially launched the construction process. The primary focus lies in establishing a comprehensive management system based on the digital foundation platform of a smart campus. This will enable the integration and optimization of the school's information value chain, providing strong technical support for its overall development. Centered around the digital foundation at the school level and with educational and teaching model reforms as its core, the construction of the informatization benchmark school has been steadily advancing.

At present, our big data precision education system covers all 44 classes in the school and serves over 1,900 students and teachers, with more than five years of data accumulated. Specifically, in terms of teaching applications, we aim to make breakthroughs in real-time data collection of procedural teaching processes, achieving both personalized teaching and highlighting the students' individual learning and self-directed learning. We have observed four primary outcomes:

The system collects and integrates student homework and assessment data comprehensively. This allows us to create precise personal and class academic profiles, enabling teachers to make data-driven decisions for lesson preparation, teaching delivery, and post-lesson tutoring. This promotes personalized instruction for each student.

The system's intelligent grading technology and analytical engines significantly reduce repetitive tasks for teachers. This frees up their time to focus more on emotional interactions with students and provide personalized guidance. Furthermore, it allows teachers to identify students' weaknesses in specific knowledge areas, enabling students to plan their own study paths, ultimately achieving improved efficiency while reducing workload.

The system generates reports across various dimensions—single-session, cumulative, class-level, grade-level, and school-level—allowing educators to discover teaching issues and understand their underlying causes. These reports provide practical and scientific decision-making support for teaching management within the school.

With intelligent engine-driven data, we support personalized education with a focus on "what is taught" and "what is learned." This method emphasizes mutual discussions, exchanges, and equal interactions between teachers and students. This has led to innovative teaching models and philosophies in our school, such as "theme-based interaction," enabling students to develop knowledge through intellectual exchanges while returning to the true essence of education.

Host:

During the pandemic, many schools transitioned to online education. Do you think online education can meet the requirements of smart education?

Li Wei:

We all know that the COVID-19 pandemic posed new challenges for education. In 2022, the Shanghai Municipal Education Commission temporarily required all middle and primary schools in the city to transition to online teaching. Ensuring smooth online learning for all 3 million students and teachers involved navigating a range of issues, such as network connections, platform maintenance, and real-time teacher-student interactions.

Compared to the "suspended classes but continued learning" initiative of 2020, the full-time online teaching during this pandemic period offered more valuable lessons. There were also changes in the application scenarios for livestreaming. For instance, aside from the fully-online model where both teachers and students were at home, there were cases where teachers were at home and students were in school, or where individual students at home needed real-time online access to live classrooms. These setups were supported by additional hardware, such as high-definition cameras.

On the software side, the Shanghai Municipal Education Commission launched the "Shanghai Weixiao" platform, a large-scale intelligent learning system with multiple modules such as "Air Classroom," "Face-to-Face with Expert Teachers," and "The Three Assistants." The Air Classroom module features lectures designed and taught by Shanghai's top educators, covering all grades and subjects in primary and secondary education.

In 2022, I personally participated in creating three high school information technology lessons for the Air Classroom, which took nearly two years to complete—from the organization of the teaching research team to the final filming. Expert educators refined the lessons multiple times, resulting in high-quality content that has been widely recognized and frequently used during online teaching.

Currently, many teachers apply a "dual-teacher" model during online teaching: the first 20 minutes involve playing the Air Classroom videos, and the remaining 20 minutes are used by instructors to provide supplementary explanations tailored to their students' needs. Online teaching has also led teachers to segment Air Classroom videos by knowledge points, which they use in their own lesson preparations and teaching.

The experience and models established during the pandemic will not be discarded but will gradually transition into hybrid teaching models combining online and offline methods. Beyond the platforms provided by the Shanghai Municipal Education Commission, such as "Shanghai Weixiao," many schools have built their own smart teaching platforms. For instance, our school has a Homework Practice Center platform with high user engagement. Students can submit homework using their mobile phones, enabling immediate automatic grading for objective questions and photo submissions for subjective answers.

In classrooms undergoing digital transformation trials, each student is provided with a smart tablet. Using internet connectivity and the city's "Three Assistants" platform, we have created an integrated online and offline environment where teachers and students can excel. Currently, three English classes at our school are piloting such practices. In the coming years, as benchmark schools continue to expand, advanced tools like deep learning platforms, interactive teaching systems, and in-class learning perception systems will be further refined. These advancements will greatly enhance the application of smart teaching.

Host: What challenges do you think schools might face in forming a good ecosystem for smart education?

Li Wei: As we know, ecosystems are not built overnight, and the same applies to a well-functioning smart education ecosystem. Various challenges may arise during its implementation. First, a smart education environment requires high-speed internet access, not only through wired networks but also through increasingly widespread 5G wireless connectivity. This increases bandwidth requirements, and these infrastructure developments demand support from departments at all levels.

Second, breaking down data silos across different teaching systems is an urgent issue. The inability to effectively share data leads to unnecessary duplication of data collection. If the advancement of smart education requires teachers and students to continuously input data, such initiatives will not succeed. Additionally, the rapid development of artificial intelligence technology can be intimidating for many teachers, leading them to avoid adopting these new tools due to their insufficient proficiency in information technology.

In comparison, students are "digital natives" who are not unfamiliar with digital learning. The key lies in improving teachers' capabilities to effectively use information technology in their teaching processes.

Host: As a senior teacher of information technology, what kinds of skills do you think information technology education can help students develop?

Li Wei: With the gradual implementation of new curriculums and new textbooks, especially the information technology curriculum standards for compulsory education, it has become clear that information technology represents an important branch of modern science and technology. It primarily studies the scientific principles, conceptual thinking, processes, and engineering implementations related to information in digital form and its applications.

The information technology curriculum aims to foster scientific spirit and technological ethics, enhance students' awareness of autonomy and control, cultivate core socialist values, establish a holistic perspective, and improve national digital literacy and skills. The balanced emphasis on both science and technology is a crucial characteristic of the information technology discipline. The core competencies nurtured through information technology education include information awareness, computational thinking, digital learning and innovation, and social responsibility related to information utilization. These four aspects are interconnected, mutually reinforcing, and collectively promote the development of students' digital literacy and skills.

Particularly, the curriculum emphasizes encouraging students to solve real-world problems or projects, guiding them through processes such as principle application, computational thinking, and digital tool usage. This helps them acquire knowledge and enhance their problem-solving abilities. The hardware and software features of smart education discussed earlier can effectively support students in planning, managing, and evaluating their work independently in a digital environment, emphasizing the role of students as active participants.

Host: Do you think the current examination requirements might conflict with the principles of smart education? If so, how can this conflict be resolved?

Li Wei: As mentioned earlier, the new curriculum standards for each subject emphasize the cultivation of core competencies. Specifically, they advocate for competency-based education, strengthening practical applications of disciplines, and promoting integrated learning. Moreover, the standards clarify that academic quality should not rely solely on test scores based on knowledge points but should be grounded in comprehensive assessments of process evaluation, outcome evaluation, and test design criteria. These requirements align closely with the philosophy of smart education, which achieves a holistic and precise evaluation of students using longitudinal (whole-process) assessments of their learning journey and horizontal evaluations of their moral, intellectual, physical, artistic, and labor-based qualities.

The examination requirements are also gradually evolving. For example, Shanghai's comprehensive evaluation admission system not only evaluates academic performance but also requires students to demonstrate subject-specific expertise, innovative potential, and well-rounded development. As a result, schools like ours aim to provide diverse

opportunities for students during their time at school. This includes building innovation labs where students can work on interdisciplinary projects or research topics, engaging in integrated learning activities that go beyond their standard curriculum.

Our school offers over 30 enrichment courses, and research topics are accessible to all students. Those with particular interests or strengths can also explore areas such as artificial intelligence or robotics programming, participating in competitions like the Youth Innovation Competition or various science and technology contests. These experiences and accomplishments serve as key references for university admissions, particularly for systems like comprehensive evaluation admissions.

Host: As we approach the end of the program, I'd like to ask one final question: What challenges do you think lie ahead for the future of smart education?

Li Wei: The development of educational technology has undergone significant transformations—from the early use of multimedia and electrified classrooms to today's practices in online learning, mobile learning, social networking, MOOCs, and virtual reality technologies. Advances in technology are reshaping the ways teachers teach and the ways students learn. Emerging technologies like artificial intelligence, blockchain, and big data are widely acknowledged as carrying transformative potential for society. Their continuous development challenges traditional concepts and creates both opportunities and challenges within the field of education. The digital transformation of education has become an essential driving force behind both the development of current education and future educational reform.

First, artificial intelligence-based smart education brings us personalized adaptive learning. In essence, adaptive learning respects the individual characteristics of each student, truly achieving tailored instruction. Second, precise teaching becomes possible through smart education, offering an efficient learning process that allows students to focus on educational design and personalized intervention, thereby providing higher-quality learning resources. Lastly, universal education can become a reality, breaking the boundaries between classes and schools. For example, the extensive resources of Shanghai's high school MOOCs program provide abundant, high-quality courses for students across all high schools in Shanghai.

However, smart education also faces numerous challenges. Beyond the challenges presented by artificial intelligence technologies, like ChatGPT, there are issues rooted in the technology itself and its applications. The development of smart education requires massive data support. Yet, with data structures varying across different applications, establishing a unified data foundation to enable data sharing remains a pressing issue. Additionally, the effectiveness of smart education will ultimately depend on the ability of teachers and students to utilize new technologies proficiently.

Host: Thank you very much, Teacher Li, for joining us in the studio today and sharing your insights on smart education. I'm sure our audience now has a deeper

understanding of the topic. Once again, thank you, Teacher Li, and we look forward to seeing everyone in the next episode. Goodbye!